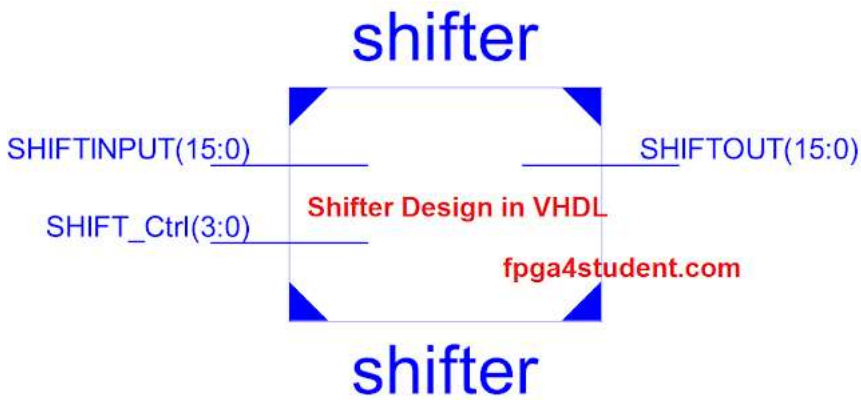
Shifter Design in VHDL

In this VHDL project, a shifter with the ability to shift and rotate data, which is mainly used in the permutation and transpositions of ciphers, will be implemented in VHDL. The VHDL shifter is a key component in the upcoming co-processor's processing unit. Fast shifting and rotating functions are critical for cryptographic applications.

VHDL code for the shifter will be presented together with its testbench VHDL code for functional simulation.



The input/ output interface of the shifter is shown in the above figure. The shifter instruction set is as follows:

- SHIFT_Ctrl = "1000": SHIFTOUT <= Rotate SHIFTINPUT >>8
- SHIFT_Ctrl = "1001": SHIFTOUT <= Rotate SHIFTINPUT >>4
- SHIFT_Ctrl = "1010": SHIFTOUT <= Shift Left Logical SHIFTINPUT << 8

VHDL code for the shifter:

```
library IEEE;
use IEEE.STD_LOGIC_1164.ALL;
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-- VHDL project: VHDL code for Shifter
entity shifter is
  generic ( N: integer:=16
  );
    Port ( SHIFTINPUT : in  STD_LOGIC_VECTOR(N-1 downto 0);
   SHIFT_Ctrl : in  STD_LOGIC_VECTOR(3 downto 0);
   SHIFTOUT: out  STD_LOGIC_VECTOR(N-1 downto 0)
  );
end shifter;

architecture Behavioral of shifter is

begin
process(SHIFTINPUT,SHIFT_Ctrl)
begin
case(SHIFT_Ctrl) is
when "1000" => SHIFTOUT <= SHIFTINPUT(7 downto 0)&SHIFTINPUT(15 downto 8);-- ROR8
when "1001" => SHIFTOUT <= SHIFTINPUT(3 downto 0)&SHIFTINPUT(15 downto 4);-- ROR4
when "1010" => SHIFTOUT <= SHIFTINPUT(7 downto 0) & "00000000"; -- SLL8
when others => SHIFTOUT <= x"0000";
end case;
end process;

end Behavioral;
```

Testbench VHDL code for the shifter:

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
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-- VHDL project: VHDL code for Shifter
-- VHDL testbench code for Shifter
ENTITY tb_shifter IS
END tb_shifter;

ARCHITECTURE behavior OF tb_shifter IS

    -- Component Declaration for Shifter
    COMPONENT shifter
    PORT(
         SHIFTINPUT : IN  std_logic_vector(15 downto 0);
         SHIFT_Ctrl : IN  std_logic_vector(3 downto 0);
         SHIFTOUT : OUT  std_logic_vector(15 downto 0)
        );
    END COMPONENT;

   --Inputs
   signal SHIFTINPUT : std_logic_vector(15 downto 0) := (others => '0');
   signal SHIFT_Ctrl : std_logic_vector(3 downto 0) := (others => '0');
  --Outputs
   signal SHIFTOUT : std_logic_vector(15 downto 0);

BEGIN

 -- Instantiate the Shifter
   uut: shifter PORT MAP (
          SHIFTINPUT => SHIFTINPUT,
          SHIFT_Ctrl => SHIFT_Ctrl,
          SHIFTOUT => SHIFTOUT
        );

   -- Stimulus process for shifter
   stim_proc: process
   begin
      SHIFTINPUT <= x"0044";
      wait for 100 ns;
  SHIFT_Ctrl <= "1000";-- ROR8
  wait for 100 ns;
  SHIFT_Ctrl <= "1001";-- ROR4
  wait for 100 ns;
  SHIFT_Ctrl <= "1010";-- SLL8
      wait;
   end process;

END;
```
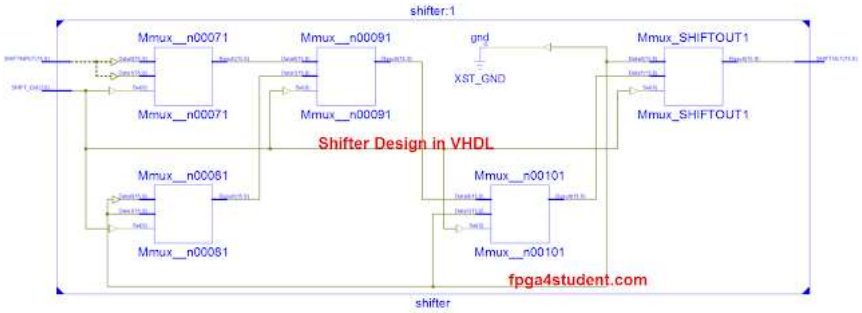

Synthesized RTL Schematic for the shifter:



Simulation Waveform for the shifter:

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